Properties of CTFT:

The properties of CTFT directly follows from the properties of Laplace Transform. We just need to take $s = j\omega$ and we are done.

Let $x_1(t) \xleftrightarrow{\mathcal{L}} X_1(s)$ ROC: R_1

$x_2(t) \xleftrightarrow{\mathcal{L}} X_2(s)$ ROC: R_2

and $x_1(t) \xleftrightarrow{\mathcal{F}} X_1(j\omega)$

$x_2(t) \xleftrightarrow{\mathcal{F}} X_2(j\omega)$

We are assuming that the Laplace transform & FT exists.

Property	Laplace Transform	FT
① Linearity $a x_1(t) + b x_2(t)$	$a X_1(s) + b X_2(s)$ ROC: $R_1 \cap R_2$	$a X_1(j\omega) + b X_2(j\omega)$
② Time shift $x_1(t - t_0)$	$e^{-st_0} X_1(s)$ ROC: R_1	$e^{-j\omega t_0} X_1(j\omega)$
③ Time scaling $x_1(at)$	$X_1(s/a) \cdot 1/\lvert a \rvert$ ROC: R_1/a	$X_1(j\omega/a) \cdot \frac{1}{\lvert a \rvert}$
④ Convolution $x_1(t) * x_2(t)$	$X_1(s) X_2(s)$ ROC: $R_1 \cap R_2$	$X_1(j\omega) X_2(j\omega)$
⑤ Differentiation $\frac{d}{dt} x(t)$	$s X_1(s)$ ROC: R_1	$j\omega X(j\omega)$

Property	Laplace transform	CTFT
⑥ Integration $\int_{-\infty}^{t} x_1(t)\,dt$	$X_1(s)/s$ ROC: R_1	$1/j\omega\, X_1(j\omega) + \pi X(0)\delta(\omega)$
⑦ Conjugation $x_1^*(t)$	$X_1^*(s^*)$ ROC: R_1^*	$X_1^*(-j\omega)$
⑧ Differentiation in frequency domain $t\,x_1(t)$	$-\frac{d}{ds} X_1(s)$ ROC: R_1	$j\frac{d}{d\omega} X_1(j\omega)$

We look into the other properties of CTFT now.

⑨ Duality —

Recall that the Fourier transform equation is

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

and the inverse Fourier transform equation is

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, e^{+j\omega t}\, d\omega$$

Both these equations looks similar, but are not identical. There is a symmetry in the equations and this leads to the idea of duality in Fourier transform.

Let us look into this property with the help of an example.

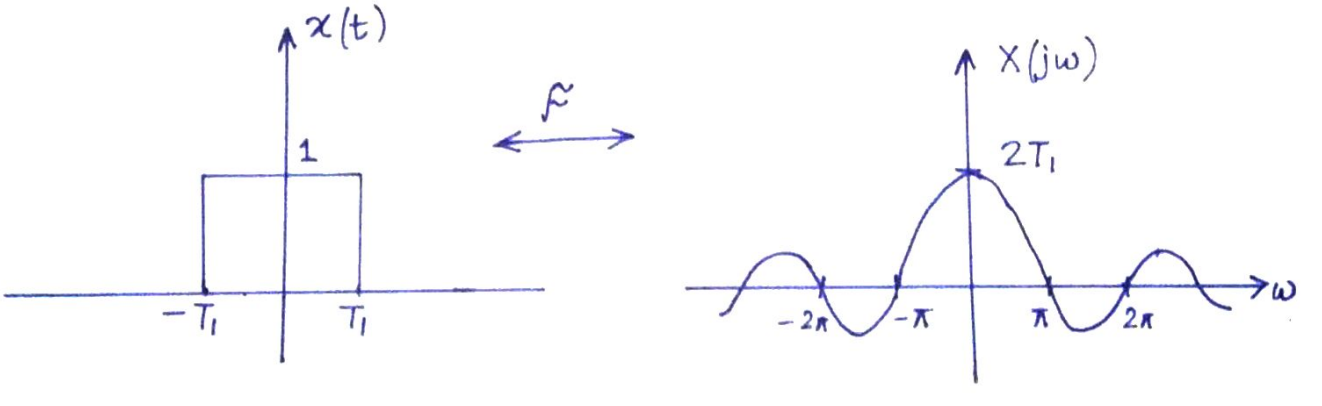
<u>Example A:</u>

Consider the rectangular pulse signal

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1. \end{cases}$$

The Fourier transform is —

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt = \int_{-T_1}^{T_1} e^{-j\omega t}\, dt = 2\frac{\sin \omega T_1}{\omega}.$$



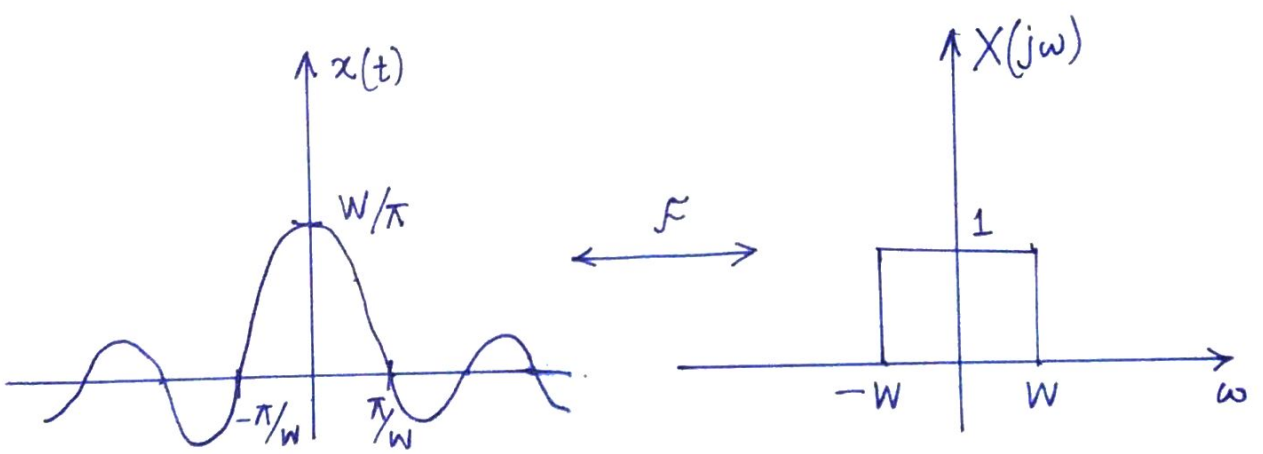
Example B: Consider a signal $x(t)$ whose Fourier transform is

$$X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W. \end{cases}$$

The inverse-Fourier transfom is

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega = \frac{1}{2\pi}\int_{-W}^{W} e^{j\omega t}\, d\omega$$

$$= \frac{\sin Wt}{\pi t}$$



Notice that in example A we had a rectangular pulse as a signal and the resulting Fourier transform was some sort of a damped sinusoidal signal. Such signals are called "sinc signal".

So, a rectangular pulse's fourier transform is a sinc signal.

And from Example B, a sinc signal's fourier transform is a rectangular pulse.

$\therefore$ We say that rectangular pulse and sinc signal are "dual" of each other.

This duality exists in fourier transform due to the symmetry in the expression of the Fourier transform and its inverse.

The duality property can be used to suggest other properties of Fourier transform. For example, we saw that differentiation in time domain corresponds to multiplication by $j\omega$ in the frequency domain. From duality, we might then suspect that multiplication of the signal by jt in the time

domain will roughly correspond to differentiation in the frequency domain.

Check property (8), this is true. Ofcourse the scaling factors are different but what is important here is that we get a rough idea about the signal.

<u>Side note:</u> A sinc function is defined as follows-

$$\text{sinc}(x) = \begin{cases} 1 & \text{for } x = 0 \\ \frac{\sin x}{x} & \text{for } x \neq 0. \end{cases}$$

(10) <u>Parseval's Relation:</u>

If $x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$, then

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega \longrightarrow (A)$$

<u>Proof:</u> $\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} x(t)\, x^*(t)\, dt$

$$\therefore \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} x(t) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(j\omega) e^{-j\omega t} d\omega \right] dt$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} x(t) \left[\int_{-\infty}^{\infty} X^*(j\omega) e^{-j\omega t} d\omega \right] dt$$

Reversing the order of integration,

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(j\omega) \left[\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right] d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(j\omega) X(j\omega) d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega \quad \#$$

The term on the left-hand side of equation (A) is the total energy in the signal $x(t)$. Parseval's relation states that this total energy may be determined either by computing the energy per unit time $(|x(t)|^2)$ and integrating over time or by computing the energy per unit frequency $(|X(j\omega)|^2/2\pi)$ and integrating over frequency.

For this reason, $|X(j\omega)|^2$ is often referred to as the energy-density spectrum of the signal $x(t)$.

$\therefore$ The Parseval's relation for finite-energy signals is the direct counterpart of the Parseval's relation for periodic signals.

(11) Multiplication Property:

We have seen that convolution in the time domain results in multiplication in the frequency domain.

So, by duality we expect that multiplication in the time domain will result in convolution in the frequency domain.

Specifically,

$$x_1(t)\, x_2(t) \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(j\theta)\, X_2(j\omega - j\theta)\, d\theta$$

$$\parallel$$

$$\frac{1}{2\pi}\left[X_1(j\omega) * X_2(j\omega)\right]$$

<u>Example:</u> Let us try to find the Fourier transform of

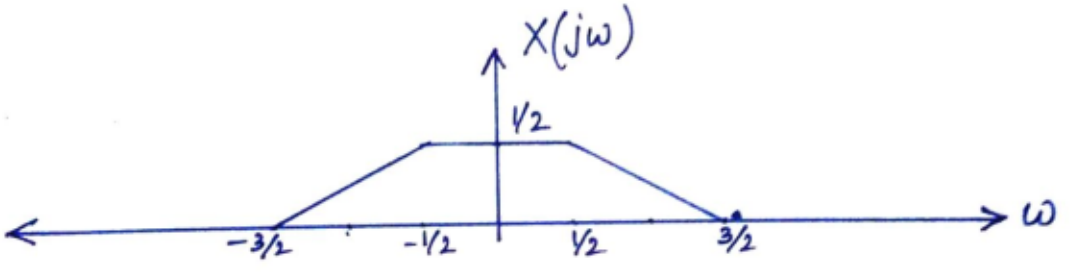
$$x(t) = \frac{\sin(t)\,\sin(t/2)}{\pi t^2}$$

Notice that

$$x(t) = \pi \cdot \left(\frac{\sin t}{\pi t}\right) \cdot \left(\frac{\sin(t/2)}{\pi t}\right)$$

Using the solution of Example B and property of multiplication, we get

$$X(j\omega) = \pi \times \frac{1}{2\pi}\, \mathcal{F}\left(\frac{\sin t}{\pi t}\right) * \mathcal{F}\left(\frac{\sin(t/2)}{\pi t}\right)$$

$$= \frac{1}{2}\, \mathcal{F}\left(\frac{\sin t}{\pi t}\right) * \mathcal{F}\left(\frac{\sin(t/2)}{\pi t}\right)$$

$$\frac{1}{2} \times X_1(j\omega) = \begin{cases} 1, & |\omega| < 1 \\ 0, & |\omega| > 0 \end{cases} \quad * \quad X_2(j\omega) = \begin{cases} 1, & |\omega| < 1/2 \\ 0, & |\omega| > 1/2 \end{cases}$$